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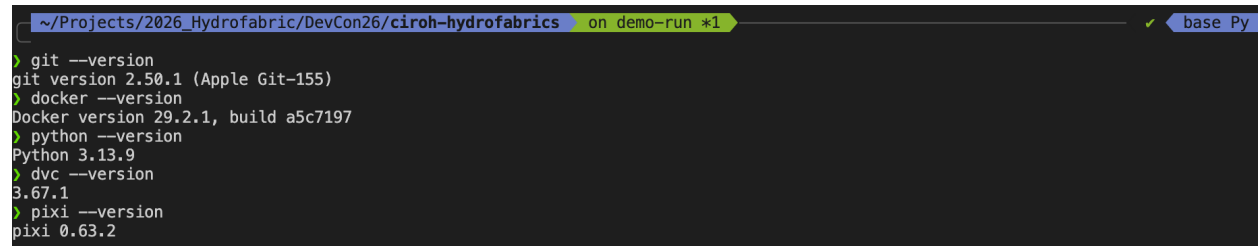
1. Before the Workshop: Required Steps for Mac

Please complete the setup steps before the workshop so that you are ready to participate in the hands-on activities.

- **Mandatory**
 - ✓ Docker Desktop
 - ✓ Python
 - ✓ Git
 - ✓ HydroShare Account
- **Optional, but recommended**
 - ✓ VS Code
 - ✓ Install Pixi
 - ✓ QGIS

2. Final Checklist

Confirm the following commands work in your terminal and ensure that you have a HydroShare account.

A terminal window with a dark background. The title bar shows the path ~/Projects/2026_Hydrofabric/DevCon26/ciroh-hydrofabrics, a green status indicator 'on demo-run *1', and a blue tab labeled 'base Py'. The terminal content shows the following commands and their outputs:

```
> git --version
git version 2.50.1 (Apple Git-155)
> docker --version
Docker version 29.2.1, build a5c7197
> python --version
Python 3.13.9
> dvc --version
3.67.1
> pixi --version
pixi 0.63.2
```

3. Detailed Installation Guidelines

Follow the instructions below for each part.

Install Python

Visit <https://www.python.org/downloads/windows/> and install the stable release.

Install Git

Visit https://git-scm.com/install/windows?utm_source and install the latest version for Windows.

Create A HydroShare Account

Visit <https://help.hydroshare.org/introduction-to-hydroshare/getting-started/create-an-account/> and create a free HydroShare account if you don't have one already.

Install Pixi

Open Ubuntu WSL terminal within VS Code and run:

```
curl -fsSL https://pixi.sh/install.sh | bash
```

```
source ~/.bashrc
```

```
pixi --version
```

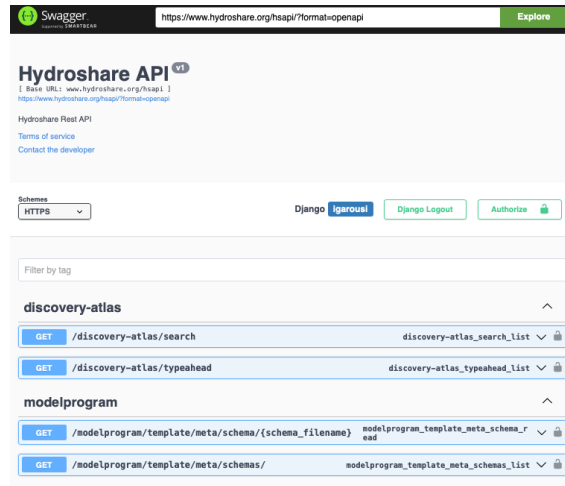
Install the QGIS

Visit <https://qgis.org/download/> and download the latest release. This is optional. You are welcome to use any GIS tool of your choice to view results of the workflow.

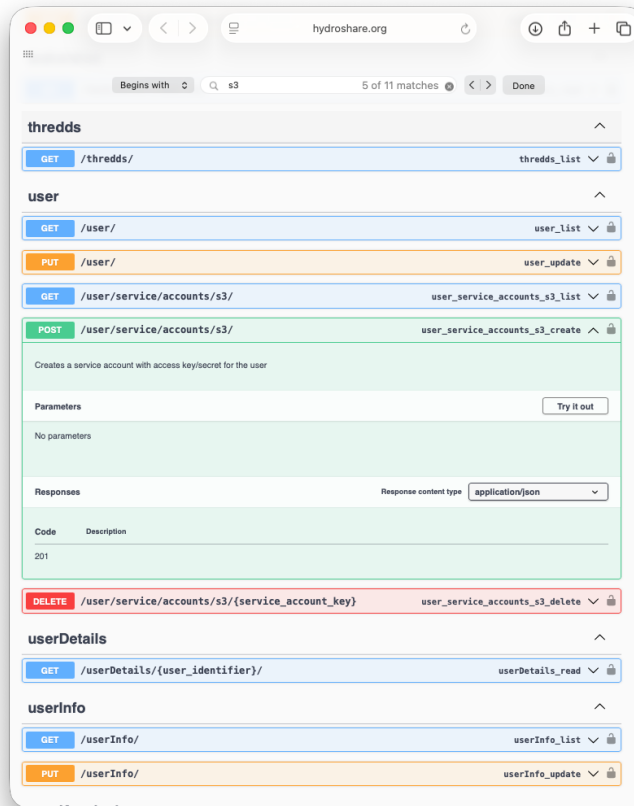
4. During the Workshop: Guided Workflow Steps

The following steps will be completed together during the workshop. They are shared in advance so you can become familiar with the workflow, expected commands, and overall process.

1. Overview of the reference and derived hydrofabrics
2. Introduce how Data Version Control connects GitHub, data, and workflow outputs
3. Run Docker Desktop and open your VS Code
4. Clone the workshop repository <https://github.com/CUAHSI/ciroh-hydrofabrics>
5. Prepare the Pixi environment. We will configure the workspace platform and install required environment (awscli & dvc). Go to the `ciroh-hydrofabrics` folder and run:
 - `pixi workspace platform add linux-64`
 - `pixi install`
 - `pixi shell`
6. HydroShare S3 Authentication for DVC
 - DVC needs access to the HydroShare-backed S3 storage where the workflow input files and cached outputs are stored. To enable this access, you need to follow these steps to configure an AWS credentials profile name `'hydrosahre'`.
 - Sign in to your HydroShare account at <https://www.hydroshare.org/>
 - Visit <https://www.hydroshare.org/hsapi/>



- Search for 'POST /user/service/account/s3/'



- Try it out and execute

POST /user/service/accounts/s3/ user_service_accounts_s3_create

Creates a service account with access key/secret for the user

Parameters Cancel

No parameters

Execute

Responses Response content type: application/json

Code	Description
201	

- Save 'access_key' and 'secret_key'

Responses Response content type: application/json

Curl

```
curl -X 'POST' \
  'https://www.hydroshare.org/api/user/service/accounts/s3/' \
  -H 'accept: application/json' \
  -H 'X-CSRF-TOKEN: zHh1KEZ7bWAOX3WSzQZKMfubcp1LND8G9Er6tghyWLIIEFmLGqWfUswxTBfwWq0' \
  -d ''
```

Request URL

https://www.hydroshare.org/api/user/service/accounts/s3/

Server response

Code	Details
201	Response body <pre>{ "access_key": "YML16LIT", "secret_key": "h0MnnpaK" }</pre> Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: https://www.hydroshare.org allow: GET,POST,HEAD,OPTIONS alt-svc: h3="443"; ma=2592000 content-language: en content-length: 93 content-type: application/json date: Thu, 21 May 2026 13:31:43 GMT server: gunicorn vary: Accept,Origin,Accept-Language,Cookie via: 1.1 google</pre> Request duration 1616 ms

Responses

Code	Description
201	

- In your terminal, run:


```
aws configure --profile hydroshare
```

```
aws configure set profile.hydroshare.endpoint_url https://s3.hydroshare.org
```
- When prompted, enter the HydroShare S3 credentials


```
AWS Access Key ID: <enter your access key>
```

```
AWS Secret Access Key: <enter your secret key>
```

```
Default region name: us-east-1
```

```
Default output format: json
```

- Inspect the project structure and key files, including workflow files, parameter files, and DVC-related metadata.
- Review the project structure and workflow components. We will walk through:
 - How the GitHub repository is organized and explain the role of the main folders and files including:
 - DVC-tracked data

- DVC remote data on HydroShare
- Workflow stages in dvc.yaml
- Workflow parameters in params.yaml
- Exact recorded outputs in dvc.lock

9. Reproduce an existing workflow state

- The goal is to demonstrate reproducibility: another researcher should be able to use the same code, data references, and same workflow definition to reproduce the same results on their own machine. We will run the workflow in a way that confirms the existing outputs can be **reconstructed** or pulled without needing to recompute everything.

```
dvc repro --pull pipelines/demo/devcon/dvc.yaml
```

```
git statu
```

```
git diff -- pipelines/demo/devcon/dvc.lock
```

10. Modify workflow parameters for the next exercise

- We will modify selected parameters in `params.yaml` that affect the resulting hydrofabric resolution. This shows how a researchers can use the same workflow to test a different configuration while keeping the process documented, tracked, and reproducible in the future.
- Visualize and compare outputs in QGIS

11. Contribute a hydrofabric correction

- In this example, we will demonstrate how a community-identified hydrofabric issue can be turned into a structured, reproducible, and transparent contribution. The goal is to show how local expert knowledge can be used to identify an issue, prepare a correction, track the associated data changes with DVC, and document the update in a way others can view and reproduce.
- Visualize and compare outputs in QGIS

12. Discuss how this supports community contribution.